

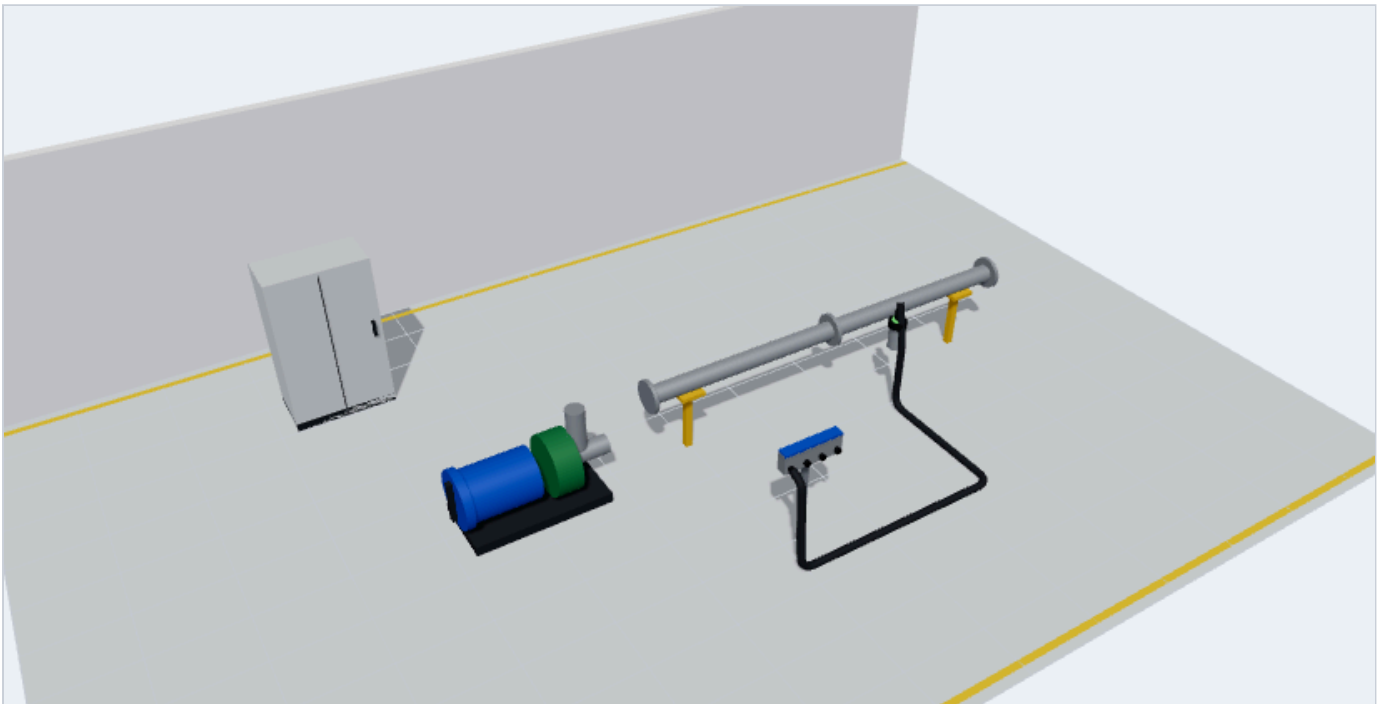
IO-Link station

IO-Link station report: commissioning data, wiring and bill of materials

STATION IO-Link station	DATE 2026-10-11	DOCUMENT Station report, rev. 1
MASTERS 1	SENSORS 1	PORTS USED 1 of 4

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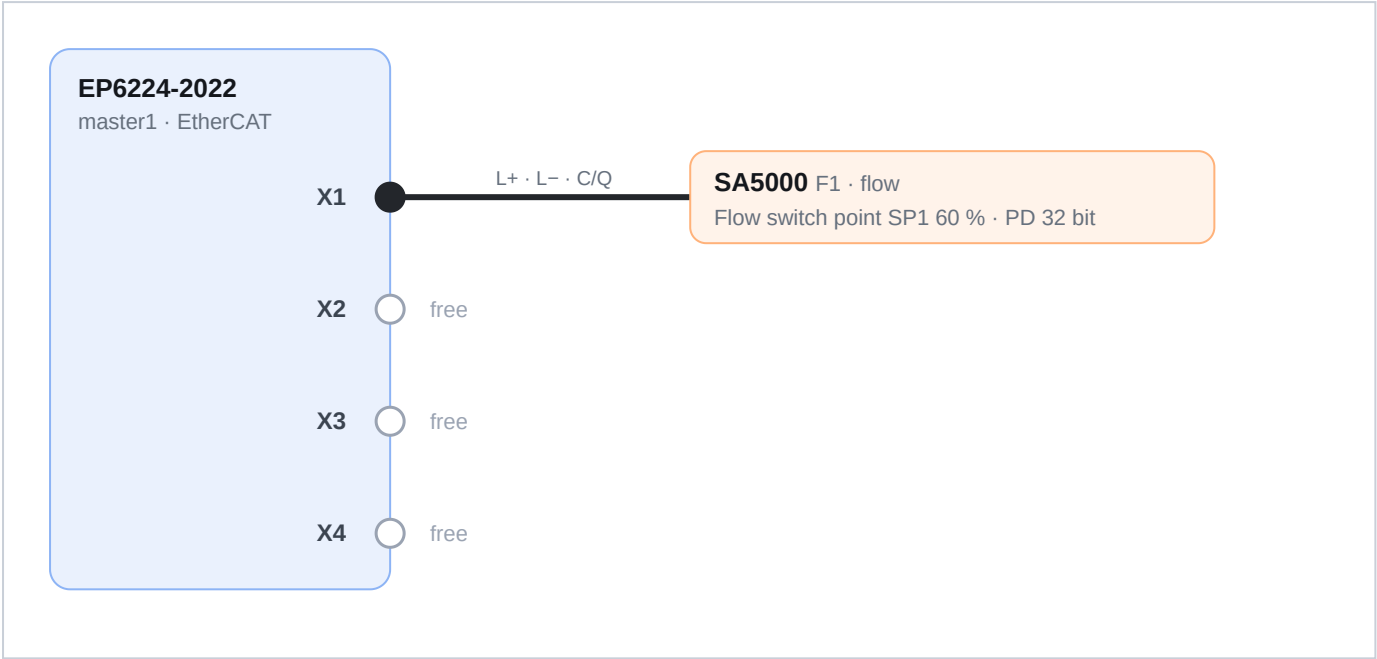
3D LAYOUT



NOTES FOR THE INSTALLER

No open points. Every sensor is inside its datasheet and wired to a master port.

1 Wiring



2 Cables

FROM	TO	DEVICE	ROUTE, M	ORDER LENGTH	CHECK
master1 X1	F1	SA5000	6.4	10 m	≤ 20 m

Route is the laid path in the 3D model; order length adds 20 % slack and rounds up to a stock length.

3 Ports and process data

PORT	DEVICE	VENDOR / DEVICE ID	CYCLE TIME	PROCESS DATA	INPUT LAYOUT	DATA STORAGE
master1 X1 IO-Link	SA5000 F1 · ifm electronic	310 / 533 type compatible	3 ms min 3 ms · COM2	4 B in 0 B out 32 / 0 bit	Flow bit 16-31, x1 % Temperature bit 2-15, x0.1 °C OUT2 bit 1 OUT1 bit 0	Backup + restore
master1 X2	Deactivated, no device wired					
master1 X3	Deactivated, no device wired					
master1 X4	Deactivated, no device wired					

Validation is set to type compatible (vendor ID and device ID must match). Use identical only when the device serial number is read at commissioning. Bit offsets count from the least significant bit of the input process data.

4 Power budget

MASTER	SENSORS DRAW	MASTER SUPPLIES	PER PORT	RESULT
master1 EP6224-2022	100 mA	4000 mA	X1 SA5000 100 mA	within limits

5 Parameter writes

PORT	PARAMETER	VALUE	ISDU INDEX.SUB	RAW	HEX	TYPE	STATUS
master1 X1 F1	Flow switch point SP1	60 %	583.0	60	00 3C	IntegerT	ok
master1 X1 F1	Flow reset point rP1	50 %	584.0	50	00 32	IntegerT	ok
master1 X1 F1	Operating mode	relative	-	-	-	-	no index

DATA GAPS

- F1 Operating mode: ISDU index not filed, set it in the parameterisation tool.

Raw = round((value - offset) / gradient). Write with the master's ISDU function block or import the commissioning file. Values for inductive sensors are already converted to the steel-equivalent switching distance.

6 Bill of materials

QTY	KIND	VENDOR	PART	VENDOR / DEVICE ID
1	master	Beckhoff	EP6224-2022	
1	sensor	ifm electronic	SA5000	310 / 533
1	cable		M12 A-coded 3/4-pole cable, 10 m	

7 Installer checklist

- ☐ Use M12 A-coded cable, straight-through, with L+ (pin 1), L- (pin 3) and C/Q (pin 4) connected. IO-Link allows 20 m per port.
- ☐ Set each master port to IO-Link and validate vendor ID and device ID before writing parameters.
- ☐ Switch data storage to backup and restore where listed, then write the parameters once and read them back.
- ☐ Do not change the master cycle time below the device minimum.

COMMISSIONED BY		DATE	
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